

Water balance & flood risk modeling

What this tool produces · *A hydrological simulation report detailing surface runoff volumes, drainage capacity thresholds, and the mapped efficacy of specific Low-Impact Development (LID) features.*

G2 · Proof of Fit

G4 · Proof of Sustainability

Dimensions

Environmental

Spatial

Preparing the workshop · *Water balance & flood risk modeling*

G2 · Proof of Fit

Why this tool matters

Reconnecting the paved, rigid city with the ancient, fluid logic of water, allowing rain to be absorbed as a life-giving resource rather than expelled as a hazardous nuisance.

Evidence we produce

A hydrological simulation report detailing surface runoff volumes, drainage capacity thresholds, and the mapped efficacy of specific Low-Impact Development (LID) features.

Duration & Material

Duration · 4 to 8 weeks

Material · Storm Water Management Model (SWMM) software, meteorological rainfall data, GIS elevation models

Reference case

Urban planners utilize tools like the EPA's Storm Water Management Model (SWMM) coupled with multiple attribute decision-making to optimize the placement of green infrastructure. In dense urban contexts, simulating scenarios with permeable pavements and localized bioswales allows cities to accurately forecast the reduction in peak flood volumes and prevent combined sewer overflows during extreme climate-induced rain events.

Suggested agenda · Brief (10 min) > Method walk-through (15 min) > Animation canvas (60-90 min) > Harvest (20 min)

Hydrological cycle × flood risk × resilience

Duration
4 to 8 weeks

Inputs

rainfall · snowmelt · groundwater

.....

.....

.....

Storage

soils · aquifers · wetlands

.....

.....

.....

Flows

streams · sewers · runoff

.....

.....

.....

Outputs

evapotranspiration · discharge

.....

.....

.....

Flood risk

pluvial · fluvial · coastal

.....

.....

.....

Resilience

NbS · grey infra · early warning

.....

.....

.....

Harvesting our *learnings*

After running the canvas, capture three layers of insight.

What inspires me

Stand-out signals, surprising stories, ideas to remember

How to apply it here

Concrete adaptations to our context, our team, our constraints

What we need to be autonomous

Skills, allies, resources, decisions still to secure

Each participant fills a copy. Then debrief together to surface patterns and next steps.